

PHENOVA — MASTER PROGRAM SPECIFICATION

Final Hardened Competitive Edition • Position A Independent Operation + Automatic IFEC Enhancement

Unified program combining the hardened production foundation with the supplied Competitive Edition capability expansion. The Competitive Edition retains the foundational binding rules and adds competitive capabilities across AI direction, transcript editing, media intelligence, professional editing, audio intelligence, reframing, compositing, future generation, collaboration, transparency, and reliability.

0. Binding Master Directive

- Build PHENOVA as a real production-capable AI video editor. Compilation, APK generation, navigation, API existence, JSON changes, or simulated progress are never acceptance.
- Never use fake AI, simulated rendering, dummy exports, fake persistence/authentication/Premium, fabricated media metadata, placeholder controls, or production development endpoints.
- This program combines the hardened PHENOVA foundation with the Competitive Edition expansion. Existing security, quota, architecture, testing, IFEC, persistence, and Definition-of-Done rules remain binding.

1. Product Identity

- PHENOVA; Android-first, mobile-first, iOS-ready architecture.
- AI-assisted video editing with genuine media processing for ordinary users through progressively professional workflows.
- PHENOVA must function independently of IFEC and future external infrastructure.

2. Independent Operation + Automatic IFEC Enhancement

- PHENOVA supplies its own core editing, media processing, projects, storage, authentication, backend, rendering, AI fallback, quotas, and subscriptions.
- When registered on IFEC, IFEC automatically becomes an enhancement layer for AI Gateway/model routing, specialist models, agents, heavy rendering, optional synchronization, metering, diagnostics, testing, and ecosystem services.
- If IFEC is unavailable, timed out, or errors, PHENOVA automatically falls back to its own capabilities. No core feature may permanently break because IFEC is offline.
- User-facing state may indicate Enhanced by IFEC / Running independently without exposing internal architecture.

3. Canonical Project System

- AI Director, transcript, timeline, creative tools, professional controls, media intelligence, and project history are multiple interfaces over one canonical project state.
- Manual editing, transcript editing, AI editing, corrections, and rendering must use the same project representation and validated Edit Plan/operation system.
- Project state is versioned and auditable.

4. Media Import + Validation

- Import multiple videos, photos where appropriate, and audio/music through proper device gallery/media selection.
- Extract real duration, dimensions, FPS, codec, audio streams, file size, container metadata, and other supported metadata.
- Validate actual media content, not extensions alone. Reject corrupt, unsupported, oversized, malicious, or mismatched files.
- Protect ownership/path boundaries and support resumable/chunked uploads.

5. Core Editing Engine

- Real trim, cut, split, merge, reorder, delete, crop, resize, aspect ratio, speed, slow motion, reverse where supported, transitions, text, captions/subtitles, colour adjustment, volume, mute, audio mixing, and music synchronization.
- Use FFmpeg or equivalent production-grade media processing. Every operation must affect actual rendered media.
- Maintain a versioned capability registry; unsupported operations must never appear as working features.

6. Audio Intelligence

- Use supplied audio according to instructions; otherwise retain original clip audio.
- Support keep/mute/volume/mix/dominant-source controls, plus speech enhancement, noise/echo cleanup, silence/filler removal, room-tone handling, normalization, ducking, beat analysis, and intelligent music placement where technically feasible.
- Audio decisions must preserve dialogue intelligibility and remain auditable.

7. AI Director / Editing Agent

- Plan and execute multi-step edits from natural-language instructions.
- Inspect actual project state, media analysis, timeline, capabilities, constraints, and intent before acting.
- Create complete first cuts, revise them, explain decisions, and make targeted changes without unnecessarily rebuilding unrelated work.
- Support automatic, guided, and manual modes over the same project.

8. Strict Edit Plan + Operation Graph

- Every AI action produces a strict, versioned, machine-readable Edit Plan.
- Validate schema, media, capability registry, entitlements, safety, and feasibility before execution.
- Support grouped/conditional operations while retaining deterministic execution.
- Maintain an operation graph/edit log recording affected assets, operations, reasons, confidence where available, and rollback relationships.
- Invalid, unsafe, impossible, unsupported, or unauthorized plans must be rejected.

9. Story + Creative Intelligence

- Analyze hook, setup, development, climax/primary point, resolution/call-to-action where applicable.
- Evaluate pacing, repetition, continuity, visual variety, speaker balance, shot quality, audio continuity, and unnecessary gaps.
- Support cinematic, energetic, documentary, educational, promotional, vlog, interview, social-short, and user-defined creative goals.
- Offer alternatives when multiple reasonable interpretations exist and distinguish objective corrections from subjective choices.

10. Unified Media Understanding

- Build a cached multimodal project index for supported transcript/speech, faces, speakers, objects, scenes, shots, camera motion, silence, audio, visual quality, metadata, and timeline context.
- Support natural-language search for people, objects, topics, locations, phrases, reactions, and visual events where implemented.
- Never claim analysis that was not actually performed. Cache expensive analysis.

11. Dual Editing Surface

- Provide conventional timeline and transcript/document editing over the same project state.
- Edits in either surface remain synchronized. AI actions modify the same canonical representation.
- Users can move from automation to frame-level/manual precision.

12. Professional Editing Depth

- Progressively add multi-track editing, keyframes, masks, motion tracking, blend modes, advanced transitions, advanced colour, professional audio, compositing, animation, bins, folders, tags, labels, markers, searchable metadata, and large-project media management.
- Every capability must be real, versioned, tested, and gated by the capability registry.

13. Smart Reframing + Multi-Output

- Automatically reframe subjects for vertical, landscape, square, and other outputs.
- Generate platform-specific variants from one canonical project.
- Support safe areas, subject priority, caption regions, and brand layout constraints. Test subject preservation and output dimensions.

14. Creative Effects + Compositing

- Real background removal/replacement, masking, tracking, overlays, controlled restyling, motion graphics, and effect stacks when implemented.
- AI selects effects according to the creative brief and keeps parameters editable/removable.

15. Generative Editing — Future Layer

- Provider-neutral interfaces for future video/image/audio generation.
- Generated assets enter projects as identifiable, editable assets with provenance and distinction from originals.
- Support generative extension/repair when a compatible provider exists.
- Creative Generation is not a current dependency; this is an architectural provision for later activation.

16. Projects + Version History

- Create, rename, save, reopen, delete, continue, and search projects; show recent projects and AI conversations.

- Persist across app/backend restarts.
- Default version policy: retain the latest 20 versions per project, configurable, and prune only versions not required by active jobs.
- Support restore, targeted undo, global undo/redo, and reversible AI operation groups. Optional IFEC sync.

17. Corrections + Last-Mile Control

- Natural-language corrections modify the existing project/Edit Plan without restarting.
- Preserve unrelated edits where possible.
- Support targeted accept/reject/undo. Corrections must affect real rendered media.

18. Authentication + Ownership

- Real email/password authentication, secure hashing, persistent sessions, secure logout/revocation, account recovery, and Google sign-in architecture.
- User-specific projects, media, settings, versions, quotas, and subscriptions.
- Every protected resource enforces server-side ownership.

19. Free Plan

- First month: 1 Pro edit/day; first 3 days 1080p; remaining first month 720p; 2 Basic edits/day from the beginning.
- After month one: 2 Basic edits/day at 720p; 5 corrections per completed free edit.
- Basic: max 5 clips, max 5 minutes/clip OR one video up to 30 minutes.
- Daily reset: 00:00 UTC based on server time.
- Quota rules: accepted distinct edit jobs consume edits; pre-acceptance validation failures and automatic infrastructure retries do not; user-requested new edits do. Accepted corrections consume correction allowance; infrastructure retries do not. Same-state re-renders do not consume new edit quota unless they represent a new user-requested edit.
- All rules are server-enforced and tested.

20. Premium + Pricing

- Premium: max 15 clips, max 5 minutes/clip OR one video up to 2 hours, 1080p, no daily one-edit restriction, unlimited corrections.
- Pricing: 3-day ■1,500/\$1; monthly ■15,000/\$10; yearly ■90,000 with configurable USD equivalent.
- Centralized configurable pricing; server-side purchase verification; payment-provider abstraction.

21. Backend + Database

- Production API for authentication, users, sessions, projects, versions, media, uploads, Edit Plans, AI requests, rendering jobs, exports, corrections, subscriptions, quotas, notifications, and errors.
- PostgreSQL/equivalent persistent database with migrations. Separate development/staging/production configuration.

22. Rendering + Job System

- FFmpeg/equivalent. Job states: QUEUED, PROCESSING, COMPLETED, FAILED, CANCELLED.
- Progress, cancellation, retry, idempotency, output validation, cleanup, and failure reporting.
- Hybrid local/PHENOVA backend/IFEC rendering.
- Memory-aware processing, bounded concurrency, streaming/chunking, backpressure, and thermal-aware reduction/deferment of local rendering where platform APIs permit.
- Prevent uncontrolled storage growth.

23. Offline + Degraded Behaviour

- Local core editing, preview, and previously rendered projects remain available offline where technically possible.
- Cloud AI/rendering fails gracefully and retries where appropriate.
- IFEC failure triggers PHENOVA fallback. Never fabricate analysis, edits, or success.

24. Storage + Media Lifecycle

- Storage abstraction for local development and S3-compatible/object storage in production.
- Per-user and per-project quotas; resumable uploads; cleanup of abandoned uploads, failed jobs, temporary files, obsolete outputs, and deleted-project media.

25. Security + Production Hardening

- Proper release signing; no debug signing. Disable unnecessary production cleartext HTTP.
- Protect secrets/credentials; secure temporary media; validate uploads; server authorization; rate limiting; secure CORS; certificate pinning recommended.
- Never ship 10.0.2.2:8000 or any development endpoint in production. Never expose developer/IFEC internals to ordinary users.

26. Android

- Proper gallery/media selection, multiple videos/audio, permissions, large-file/lifecycle handling, restart resilience, production API configuration, release APK/AAB, and real-device testing.
- Design for low-memory devices and sustained media-processing workloads.

27. Collaboration + Review — Future

- Optional sharing, comments, review links, approvals, collaborator roles, and team workflows.
- Immutable versions protect prior work. Review notes can become structured AI correction requests.

28. AI Transparency + Provenance

- AI edit log; accept/reject where appropriate; targeted undo/restore; capability limitations before execution where possible.
- Never hide AI failure behind successful-looking render. Retain generated-asset provenance and provider/model metadata internally where available.

29. Provider Architecture + Reliability

- AI, media analysis, rendering, and future generation providers are replaceable through interfaces.
- Provider failure triggers retry/fallback according to policy. Projects remain editable if optional providers disappear.
- Long-running work uses job state, progress, retry, cancellation, and output validation.

30. Testing

- Automated authentication, projects, ownership, media, editing, audio, rendering, export, quotas, billing, APIs, errors, retries, persistence, security, and IFEC fallback.
- Real FFmpeg tests. AI first-cut tests with real media. Correction stability tests. Transcript/timeline synchronization tests. Media-intelligence retrieval tests. Pacing/continuity tests. Multi-aspect reframing tests. AI audit/reversibility tests. Provider-failure tests. Generated-asset provenance tests when generation exists.
- Large-project performance tests must cover storage, indexing, timeline responsiveness, memory, thermal behavior, render concurrency, and recovery.
- Regression tests must protect no-fake-output, quota, security, ownership, persistence, and Definition-of-Done rules.

31. End-to-End Acceptance

- Install release APK → account → sign in → import multiple videos/audio → verify original audio → create project → inspect media → AI prompt → validated Edit Plan → real edits → actual preview → render → export → save → restart → reopen → correction → render → export again.
- Repeat with IFEC available and completely unreachable. Test invalid media, network failure, quotas, Premium, logout/login, restart, low memory, thermal pressure, provider failures, retries, and recovery on a real device.

32. Build + Deployment + Deliverables

- Provide dev/staging/production templates, .env.example, database schema/migrations, API docs, deployment docs, startup/migration commands, signing instructions, APK/AAB instructions, README, test suite, and reproducible release process.
- Final repository: Flutter source, backend, Android configuration, database, media engine, AI service, Edit Plan schema, rendering/jobs, storage, authentication, subscription system, IFEC adapter, tests, docs, and release configuration.
- Production builds must fail clearly when required production configuration is absent.

33. Previous Failure Corrections — Mandatory

- Every visible button must work. No fake rendering/export, fake AI, fabricated metadata, fake corrections, fake Premium activation, fake persistence, or UI-only feature.
- Proper gallery/media selection. Real persistent storage. No debug signing. No production cleartext HTTP. No production 10.0.2.2:8000 fallback. No developer/IFEC internals exposed.
- Build completion is never product completion.

34. Implementation Priority

- Retain production foundation phases: repository/configuration → backend/database/auth → real media/storage → rendering/jobs → core editor → AI/Edit Plan → subscriptions/security → offline behavior → testing/deployment.
- Competitive A: media understanding + canonical project model. B: AI Director/first-cut + operation log. C: transcript/document editing. D: professional timeline/audio/colour/compositing/tracking/capability registry. E: smart reframing/multi-output. F: collaboration/review. G: provider-neutral generative editing when Creative Generation becomes active.
- Every phase requires implementation + UI + backend integration + persistence + errors + security + tests + real execution verification.

35. Definition of Done

- A feature is complete only when implementation, UI connection, required backend integration, persistence, error handling, security, tests, and real execution verification are complete.
- Buttons, screens, APIs, JSON fields, model calls, or compilation do not prove functionality.
- If genuine implementation is not available, mark it incomplete and do not present it as working.

36. Final Product Principle

- PHENOVA combines AI Director, natural language, transcript, timeline, creative tools, professional controls, media intelligence, and reversible history over one canonical project system and one validated operation system.
- AI must be powerful, auditable, reversible, constrained by real capabilities, and incapable of fabricating successful work.

37. Final Acceptance Criterion

- A real user can install PHENOVA, create an account, import real clips/audio, request edits naturally, have PHENOVA inspect actual media, create and validate an executable Edit Plan, perform genuine edits, render a genuine video, preview it, export it, save/reopen the project, make corrections, render again, and export again.
- This must work independently of IFEC and with automatic IFEC enhancement when available. The finished product is a functioning AI video editor, not a UI prototype.